Quantum Neuroeconomics: Finite Mathematical Optimization in Decision Neurobiology

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Abstract

This article presents a framework integrating finite mathematics and quantum machine learning to optimize neuroeconomic decision-making processes. By modeling human portfolio selections as finite Markov decision processes, we reveal the computational foundations of choice under uncertainty.

1. Introduction

Neuroeconomics seeks to understand how the brain computes value and drives behavior. Classical models often fail to capture the discrete, finite nature of synaptic computations. Here, we apply combinatorial optimization and finite probability to simulate and predict portfolio choices in dynamic markets.

2. Finite Mathematics in Choice Computations

A subject's choice among discrete alternatives forms a finite state space, S. Expected utility is maximized over a predefined combinatorial set:

$$E[U] = \sum_{s=1}^{S} P(s)U(x_s)$$

Linear constraints limit the resources (e.g., cognitive capacity, capital):

$$\sum_{i=1}^{n} c_i x_i \leq C, \quad x_i \in \{0, 1\}$$

3. Quantum Machine Learning for Neuro-Portfolios

We model firing rates as probability amplitudes in a quantum circuit. Superposition represents the deliberation phase before state collapse (choice execution):

$$|\Psi\rangle = \alpha|Buy\rangle + \beta|Sell\rangle$$

4. Conclusion

Coupling quantum algorithms with finite mathematics elucidates the mechanics of neuronal decision-making, offering scalable telemetry for financial modeling.